

$$\frac{1}{y^2-4} \frac{dy}{dx} = 1$$

$$y=2 \quad y=-2 \quad \left. \begin{array}{l} \text{Problem} = \frac{1}{0} \frac{dy}{dx} \end{array} \right\} \begin{array}{l} 2, -2 \text{ are singular} \\ \text{solutions} \end{array}$$

$$\int \frac{1}{y^2-4} dy = \int 1 dx$$

$$\int \frac{1}{(y-2)(y+2)} dy = x + C$$

$$\int \frac{A}{y-2} dy + \int \frac{B}{y+2} dy = x + C$$

$$\int \frac{-1/4}{y-2} dy + \int \frac{1/4}{y+2} dy = x + C$$

$$\frac{1}{4} (\ln(y-2) - \ln(y+2)) = x + C$$

$$\ln(y-2) - \ln(y+2) = 4x + C$$

$$\ln\left(\frac{y-2}{y+2}\right) = 4x + C$$

$$\frac{y-2}{y+2} = e^{4x+C}$$

$$y-2 = e^{4x+C} (y+2) \rightarrow y-2 = ye^{4x+C} + 2e^{4x+C}$$

$$y - ye^{4x+C} = 2e^{4x+C} + 2$$

$$y = \frac{2e^{4x+C} + 2}{1 - e^{4x+C}} = y = \frac{2ke^{4x} + 1}{1 - ke^{4x}}$$

